

# Memory Controllers for DDR Memories

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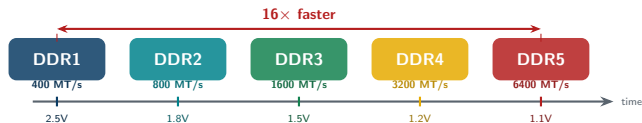
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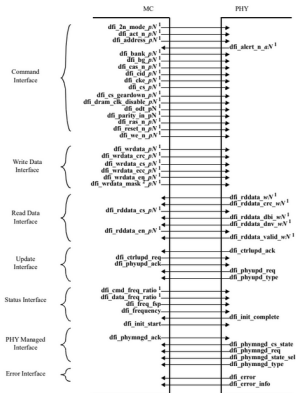
# Recap: Memory Controller for DDR Memories

- Acts as a bridge between the SoC and DRAM
- Enforces 50+ timing constraints (tWTR, tRC, tFAW...)
- Handles row-buffer management and request scheduling
- Manages power modes: PASR, TCSR, Deep Power-Down
- AXI out-of-order IDs map naturally to DDR bank parallelism



|          | DDR1 | DDR2 | DDR3 | DDR4 | DDR5 |
|----------|------|------|------|------|------|
| Prefetch | 2n   | 4n   | 8n   | 8n   | 16n  |
| Banks    | 4    | 4    | 8    | 16   | 32   |

# DFI Interface: MC ↔ PHY Signal Groups



## Command Interface

- Address, bank, CS#, CKE, ODT, RAS#, CAS#, WE#
- MC → PHY; alert feedback PHY → MC

## Write Data Interface

- `dfi_wrdata`, `dfi_wrdata_en`, `dfi_wrdata_mask`
- CRC and ECC sideband signals

## Read Data Interface

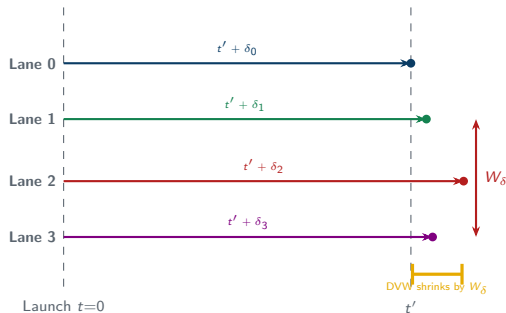
- `dfi_rddata_en` MC → PHY (look-ahead)
- `dfi_rddata`, `dfi_rddata_valid` PHY → MC

## Update / Status / PHY Managed

- Handshake: `ctrlupd`, `phyupd`, `phymngd`
- Freq ratio, init start/complete, error reporting

# Inter-Lane Skew: Off-Die Signal Integrity

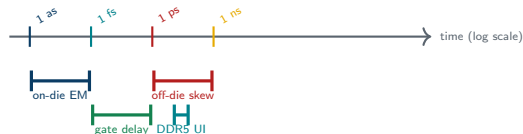
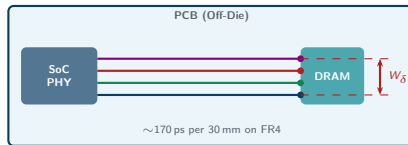
- DDR transfers data in parallel across multiple DQ byte lanes
- Each lane follows a different PCB path, picking up delay  $\delta_i$
- Skew window:  
 $W_\delta = \max(\delta_i) - \min(\delta_i)$
- DDR5 UI  $\approx 156$  ps; 20 ps skew eats >12% of it
- PHY training (Write Levelling, DQS Gating) corrects  $\delta_i$  per lane



# Off-Die vs On-Die Delay

## Off-Die (Picoseconds)

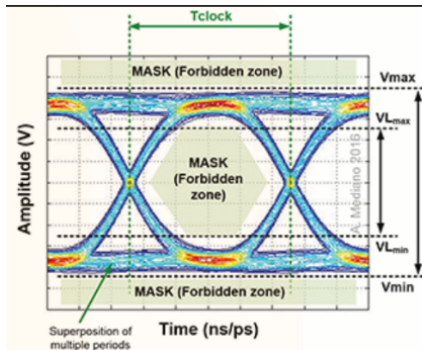
- Signals traverse PCB traces, vias
- FR4 velocity  $\approx 0.6c$ ;  $\sim 1$  ps per  $180 \mu\text{m}$
- 30 mm trace  $\approx 170$  ps;  $W_\delta$  typically 5–50 ps



## On-Die (Femto/Attoseconds)

- nm-scale on-chip metal interconnect
- $1 \mu\text{m}$  wire  $\approx 3\text{--}10$  as propagation
- Gate delay (3 nm): 1–5 ps; bottleneck is logic depth

# Eye Diagram Fundamentals



## Key Regions

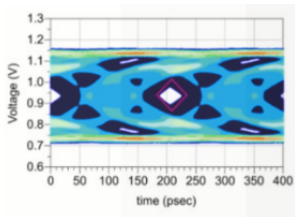
- **Eye Width:** horizontal opening between transition windows
- **Transition Window:** region where 0→1 or 1→0 switching occurs
- **Steady-State Window:** stable HIGH or LOW; sampled here
- **Unit Interval (UI):** one bit period; eye must open within it

## Why it matters

- Receiver samples at eye centre for maximum margin
- Noise and jitter reduce eye opening
- At DDR5-6400: UI = 156 ps — any skew directly closes the eye

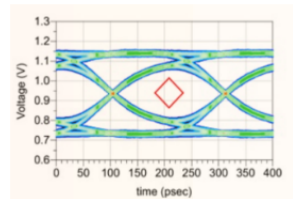
# Eye Diagram: Trained vs Untrained

Untrained Eye



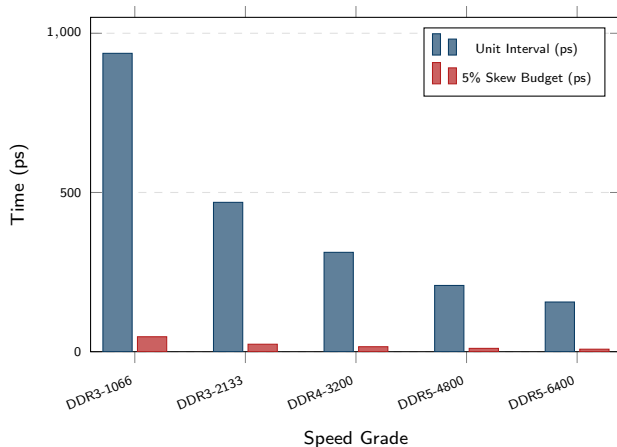
- Lane skew  $\delta_i$  closes eye horizontally
- VREF offset closes eye vertically
- ISI and jitter blur transition edges
- High BER — unreliable data capture

Trained & Calibrated Eye



- Write Levelling corrects DQS-to-CK skew  $\delta_i$
- Read centering places sample point at eye centre
- ZQ calibration sets correct driver impedance
- 2D VREF scan optimises voltage margin  $M_V$

# Why Picoseconds Matter at Modern DDR Speeds



| Speed     | UI     | 5% Budget     |
|-----------|--------|---------------|
| DDR3-1066 | 937 ps | 46.9 ps       |
| DDR3-2133 | 469 ps | 23.4 ps       |
| DDR4-3200 | 312 ps | 15.6 ps       |
| DDR5-4800 | 208 ps | 10.4 ps       |
| DDR5-6400 | 156 ps | <b>7.8 ps</b> |

- At DDR5-6400 skew budget is just **7.8 ps**
- Uncompensated off-die skew can fill this entirely
- PHY training trims  $\delta_i$  to  $\ll 5$  ps per lane



# DFI: The Standard MC-PHY Interface

- **DFI** is the industry-standard interface between MC and DDR PHY
- Decouples MC from PHY implementation details
- Covers command, data, training and power signals
- DFI 5.1: PHY-independent boot, expanded freq range, enhanced calibration feedback



| Ver        | Key Addition            | Support         |
|------------|-------------------------|-----------------|
| 1.0        | Initial spec            | DDR, DDR2       |
| 2.0        | Training interface      | DDR3, LPDDR2    |
| 3.0        | Low-power handshake     | DDR4, LPDDR4    |
| 4.0        | DDR4/LPDDR4 paths       | DDR4, LPDDR4    |
| 5.0        | PHY-independent boot    | DDR5, LPDDR5    |
| <b>5.1</b> | <b>Full back-compat</b> | <b>All gens</b> |

# DFI 5.1 Key Features

## Three Pillars of DFI 5.1

- ❶ **PHY-Independent Boot**  
MC initialises DRAM without knowing PHY internals
- ❷ **Expanded Frequency Range**  
Covers low-power LPDDR through DDR5-6400
- ❸ **Enhanced PHY↔MC Interaction**  
Real-time calibration feedback and margining

## Backward Compatibility

- Supports DDR through DDR5 and LPDDR through LPDDR5
- One DFI 5.1 MC paired with generation-specific PHY covers every standard

## Why it matters for our MC

- Single RTL targets DDR3, DDR4, DDR5 and LPDDR4
- Swapping PHY = different memory standard, same controller

# DDR3 / DDR4 / DDR5 — JEDEC Key Specs

| Parameter               | DDR3 (JESD79-3F)           | DDR4 (JESD79-4C)     | DDR5 (JESD79-5B)     |
|-------------------------|----------------------------|----------------------|----------------------|
| <b>Data Rate (MT/s)</b> | 800–2133                   | 1600–3200            | 3200–6400+           |
| <b>Prefetch / BL</b>    | 8n / BL8 (fixed/fly)       | 8n / BL8 (fixed/fly) | 16n / BL16 mandatory |
| <b>Bank Groups</b>      | None                       | 2 or 4 BGs           | 8 BGs (sub-ch.)      |
| <b>Channels</b>         | Single                     | Single               | Sub-channel ×2       |
| <b>IO Voltage</b>       | 1.5 V / 1.35 V             | 1.2 V                | 1.1 V                |
| <b>On-Die ECC</b>       | Optional                   | Optional             | Required (×4 org)    |
| <b>Refresh</b>          | tREFI = 7.8 μs             | tREFI = 7.8 μs       | Dual-rate / PBR      |
| <b>Termination</b>      | RTT_NOM                    | + RTT_PARK           | + DFITMG             |
| <b>New timing</b>       | tRCD, tRP, tRAS, tWR, tRFC | + tWTR_L/S, tRRD_L/S | + tRFC1/2/sb, tCS    |

# DDR3 JEDEC Deep-Dive

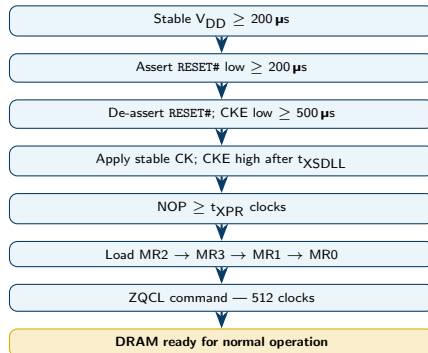
## Critical Timing Parameters

|             |                                   |
|-------------|-----------------------------------|
| <b>tRCD</b> | Row open → column command         |
| <b>tRP</b>  | Precharge → next ACT              |
| <b>tRAS</b> | Min row active time               |
| <b>tRC</b>  | tRAS + tRP (full row cycle)       |
| <b>tWR</b>  | Write → precharge recovery        |
| <b>tRFC</b> | Refresh cycle (density-dependent) |
| <b>tFAW</b> | 4-bank activation window          |
| <b>tWTR</b> | Write → read turnaround           |
| <b>CL</b>   | CAS latency (cmd → first data)    |

## Mode Registers (MRS)

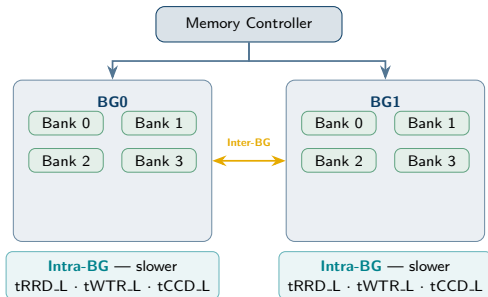
|            |                                   |
|------------|-----------------------------------|
| <b>MR0</b> | CL, BL, DLL reset, write recovery |
| <b>MR1</b> | Drive strength, RTT_NOM, AL       |
| <b>MR2</b> | CWL, ASR, SRT, RTT_WR             |
| <b>MR3</b> | MPR (multi-purpose register)      |

## Initialization Sequence



# DDR4 JEDEC Deep-Dive

## Bank Group Architecture

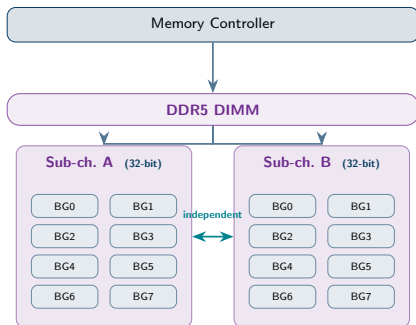


## PHY Training Procedures

- 1 **Write Levelling:** DQS launched; DRAM echoes on rising CK; adjusts DQS-to-CK skew per lane
- 2 **Read Gate:** PHY determines preamble window; gates DQS capture
- 3 **Read Centering:** Sweeps DQ VREF; finds eye centre
- 4 **Write Centering:** Tunes PHY DQ delay vs DQS
- 5 **ZQ Cal:** Periodic ZQCS; updates  $R_{ZQ}$  impedance code

# DDR5 JEDEC Deep-Dive

## Sub-Channel Architecture



## Init & Training Additions

- 1 **BCOM training:** Sideband cmd bus trained before main CA bus active
- 2 **PDA mode:** Per-DRAM addressability — tune MRS per individual device
- 3 **2× MRS sets:** Extended register space; extra latency + ODT configs
- 4 **RTT\_PARK + DFITMG:** Runtime-switchable ODT via DFI timing params
- 5 **CA parity:** DRAM validates each command frame; error flagged via ALERT\_n